

Decomposing Cross-Session Decoder Degradation in Invasive Neural Interfaces

A measurement framework for unit loss, representational drift, and gain shifts in chronic electrophysiology. Research proposal v0.1.

1. Background and motivation

Chronic invasive brain-computer interfaces (BCIs) lose decoder accuracy over days to months despite recording from the same brain region in the same subject. The phenomenon is well-documented: a decoder trained on day 1 and tested on day 30 performs substantially worse than an intra-session decoder. The BCI field has responded with drift-mitigation methods — latent dynamics models (LFADS), contrastive embeddings (CEBRA), alignment networks (Karpowicz et al., 2022), and adversarial training (Ma et al., 2023). These methods compete on aggregate cross-session accuracy, but the field lacks a principled framework for measuring *which mechanism* each method addresses.

Cross-session degradation arises from at least three distinct sources: (i) **unit loss** — units recorded on day 1 cannot be reliably tracked on day 30 due to electrode drift, tissue response, or spike-sorting failures; (ii) **representational drift** — units that are successfully tracked exhibit changed tuning to the same behavioral or stimulus variables; (iii) **gain shifts** — global changes in firing rate or response magnitude due to impedance changes, gliosis, or arousal differences. Each mechanism implies a different mitigation strategy. Better unit tracking addresses (i); latent-space alignment addresses (ii); per-session normalization addresses (iii). Without separating these contributions, the field cannot determine which methods are solving which problem.

A recent paper in mouse primary visual cortex (Hayashi et al., 2026, under review at CCN) introduced a four-condition experimental design that cleanly decomposes cross-session decoder degradation in calcium imaging recordings. They report that representational drift accounts for 59.5% of the total accuracy drop, neuron loss for 31.6%, and gain shifts for 8.9%. This decomposition has not been performed in invasive electrophysiology, where the mechanisms are likely to manifest differently — electrode drift causes more severe unit loss than calcium imaging's optical tracking, while motor cortex population dynamics may exhibit less drift than visual cortex tuning (Gallego et al., 2020).

Proposed contribution. We adapt the four-condition decomposition framework to chronic invasive recordings, leveraging the unit-matching infrastructure developed in the lab to decompose cross-session decoder degradation into its component mechanisms. The output is a measurement framework that the BCI field can adopt to evaluate drift-mitigation methods on the underlying mechanisms they address, rather than only on aggregate cross-session accuracy.

2. The decomposition framework

We define four experimental conditions, each producing one accuracy measurement. Their pairwise differences isolate the contributions of unit loss, representational drift, and gain shifts to the total cross-session accuracy drop.

Condition	Training data	Test data	Unit set	Normalization
C1	Session A (held-in trials)	Session A (held-out trials)	All units recorded in A	Session A statistics
C3	Session A (held-in trials)	Session A (held-out trials)	Only units matched to Session B	Session A statistics
C2b	Session A (full)	Session B (full)	Only units matched between A and B	Per-session (independent z-score)
C2a	Session A (full)	Session B (full)	Only units matched between A and B	Source statistics (Session A applied to B)

The decomposition identities:

C1 – C3 = Unit loss. Both conditions decode intra-session on Session A. The difference is the unit set: C1 uses every recorded unit, C3 uses only the subset that matches to Session B. The accuracy gap isolates the information

carried by units that fail to track across sessions — units that died, drifted beyond the recording radius, became poorly isolated, or were not matched by the matching algorithm. This is the cost of imperfect unit tracking, independent of any cross-session generalization.

C3 – C2b = Representational drift. Both conditions use only matched units. C3 decodes within Session A; C2b decodes across Session A → Session B with per-session independent normalization (each session's firing rates are z-scored using that session's own mean and standard deviation). Per-session normalization removes global gain differences. Any remaining accuracy drop must come from units whose *tuning* has changed: the same matched unit responds differently to the same behavioral or stimulus condition on the two days. This is representational drift in the strict sense.

C2b – C2a = Gain shifts. Both conditions decode across Session A → Session B using matched units. The only difference is normalization: C2b uses per-session statistics; C2a applies Session A's normalization parameters to Session B's data. Imposing source-session normalization re-introduces any global firing-rate or response-magnitude shifts between sessions. The accuracy gap quantifies how much degradation comes from these gain-level changes, separately from representational drift in the tuning.

Closure. By construction, the three components sum to the total intra-vs-cross degradation:

$$C1 - C2a = (C1 - C3) + (C3 - C2b) + (C2b - C2a)$$

All four accuracies are observable. The three components are computed from differences, so the decomposition is exact (no residual term). Percentages of total degradation attributed to each mechanism are well-defined and comparable across datasets, decoders, and time gaps.

3. Adaptations for invasive electrophysiology

The calcium imaging study used a discrete classification task (8 natural images) on visual cortex with $\Delta F/F$ responses as features and trial-averaged decoders. Translating to invasive BCI requires three adaptations:

Decoding task. Replace image classification with motor-cortex movement decoding. For continuous decoding (cursor velocity or position from reaching tasks), we use Kalman filters and linear ridge regressors; for discrete decoding (reach selection or BCI cursor control), we use linear discriminant analysis and small MLPs. The decomposition framework is invariant to the decoder architecture; we report the components across decoder families to characterize whether the decomposition depends on decoder choice.

Unit of analysis. Replace tracked calcium-imaged cells with tracked spike-sorted units from chronic multi-electrode recordings (Neuropixels or Utah arrays). The unit-matching pipeline becomes a first-class methodological component: rather than treating "matched" as binary, we weight each unit's contribution by match confidence (waveform similarity, drift trajectory consistency, ISI distribution preservation), and report sensitivity to matching stringency across multiple confidence thresholds.

Feature representation. Replace $\Delta F/F$ traces with spike counts in task-relevant windows (typically 100–500 ms bins relative to cue or movement onset). For continuous decoding, the features are firing rates in sliding windows; for discrete decoding, trial-averaged counts. We additionally evaluate the decomposition on latent-dynamics features (LFADS factors, CEBRA embeddings) to test whether drift contributions appear at the latent level or only at the unit level.

4. Lab infrastructure and prior work

The proposal builds on the lab's existing work in end-to-end spike localization and drift correction (Hayashi et al., NeurIPS 2026 submission). That work introduces a transformer-based Spike Localization Network (SLN) coupled to DREDge drift estimation via self-supervised alternating optimization, producing drift-corrected 3D spike localizations that are more temporally consistent than the standard monopolar baseline. The pipeline has been validated on 33 sessions across Neuropixels 1.0 and 2.0 probes spanning three multi-session datasets.

This infrastructure directly enables the decomposition project in three ways:

- **Drift-corrected localizations as input to unit matching.** Spike sorting on drift-corrected data produces more stable units across sessions than spike sorting on raw recordings. Better matches mean a more conservative C3 condition (less arbitrary loss from sorting errors) and therefore a more honest unit-loss attribution.

- **Confidence-weighted matching.** The SLN's heteroscedastic-style outputs (or analogous per-unit tracking confidence from the lab's matching pipeline) allow the decomposition to weight units by match reliability, sensitivity-testing each component across confidence thresholds.
- **Cross-session benchmarks.** The 33-session dataset already used in the lab is directly applicable. Additional public benchmarks (FALCON, IBL Brain Wide Map subsets, Churchland lab DANDI archives) provide complementary cross-validation cohorts spanning multiple species and recording technologies.

5. Experimental plan

Phase A: Decomposition on a single dataset

Establish the four-condition pipeline on one well-characterized longitudinal dataset (likely the lab's primary recordings or the FALCON benchmark). Implement all four conditions using a fixed decoder (linear ridge for continuous decoding, LDA for discrete) and the lab's current unit-matching pipeline. Report the three components and their statistical significance against permutation nulls. Validate that the components sum to the total degradation within numerical tolerance.

Phase B: Time-gap dependence

Run the decomposition at multiple inter-session intervals (1 day, 1 week, 1 month, 3 months where data permits). Quantify how each component scales with time. Hypothesis: unit loss grows fastest (electrodes drift physically); representational drift grows more gradually; gain shifts may be approximately constant or session-pair-specific. The resulting curves characterize the differential time-scaling of degradation mechanisms — a question the calcium paper raises but does not fully answer.

Phase C: Method-comparison framework

Apply standard drift-mitigation methods (LFADS, CEBRA, Karpowicz-style alignment, adversarial domain adaptation) to the same session pairs and re-run the decomposition. Each method's effect on each component becomes the headline result: *alignment methods recover X% of drift but Y% of unit loss; latent-dynamics methods recover Z% of drift but reduce unit loss only marginally*. This converts the decomposition into an evaluation framework for drift-mitigation methods.

Phase D: Robustness and sensitivity

Vary the unit-matching confidence threshold across the C3 / C2b / C2a conditions. Test whether the decomposition is robust to matching stringency. Report which components are stable across thresholds and which depend strongly on matching choices. This is the key honesty check: if "drift" changes wildly when matching parameters change, the framework needs further development before being adopted.

6. Expected outcomes

We pre-register predictions to make the framework's interpretive content explicit:

Prediction 1. Unit loss in invasive recordings will exceed the 31.6% reported in calcium imaging (estimated range: 40–60% on month-scale gaps), reflecting more severe physical drift of silicon electrodes relative to optical recording stability.

Prediction 2. Representational drift in motor cortex will be lower than the 59.5% reported in visual cortex (estimated range: 30–50%), consistent with the Gallego et al. finding that population-level dynamics in motor cortex are remarkably stable over months even when individual-unit tuning varies.

Prediction 3. Gain shifts will be comparable to or larger than the 8.9% reported in calcium imaging (estimated range: 10–25%), reflecting electrode impedance changes and gliosis-driven amplitude shifts that have no optical analogue.

Prediction 4. Latent-space alignment methods (Karpowicz-style) will recover the majority of the drift component without affecting unit loss. Latent dynamics methods (LFADS, CEBRA) will partially address both drift and gain shifts. No current method effectively addresses unit loss — this is the gap the framework will identify.

Each prediction is falsifiable by the decomposition results. Outcomes that contradict the predictions are informative findings, not failure modes.

7. Connection to broader research program

This proposal is the second instantiation of a research direction the PI has been developing: empirical decomposition of evaluation artifacts in clinical and translational neural ML. The first instantiation (Hayashi, under review at NeurIPS 2026) decomposes evaluation traps in EEG disease classification — identity leakage, lucky folds, and objective mismatch — across five cohorts and three datasets. The framework presented here applies the same intellectual approach (structured decomposition of an aggregate degradation metric into its mechanistic contributors) to a different clinical neural ML problem.

A future synthesis paper will draw together the EEG and invasive-BCI decomposition frameworks, plus a connectomics application currently under exploration with collaborators (Ratnan lab, NYU). The longer arc is a unified evaluation methodology for clinical neural ML that subsumes domain-specific failure modes under a common decomposition formalism.

8. Deliverables and timeline

Phase	Deliverable	Indicative duration
A	Four-condition pipeline implemented and validated on primary lab dataset	6–8 weeks
B	Time-gap-resolved decomposition curves across multiple inter-session intervals	4–6 weeks
C	Drift-mitigation method comparison under the decomposition framework	8–10 weeks
D	Robustness analysis and sensitivity testing	3–4 weeks
Writeup	Draft paper, internal review, submission	4–6 weeks

Total: approximately 25–34 weeks of focused work, executable across a one-year academic horizon while maintaining concurrent commitments (NeurIPS revision response on the EEG traps paper, light contributor roles on collaborator projects).

9. Target venues

Primary: NeurIPS, ICLR (BCI/neural ML tracks); eLife or Nature Communications for the cross-method comparison version with biological framing.

Secondary: Workshops at NeurIPS (Neural Latents Benchmark workshop, BCI workshop), Cosyne, Brain-Computer Interface Society conferences.

Decision criterion: deferred to Phase C. The method-comparison results determine whether the contribution is primarily ML-methodological (target ML venues) or primarily neuroscience-empirical (target neuroscience venues).

10. Key references

- Hayashi et al. (2026, under review). Representational drift dominates cross-session decoder degradation in mouse primary visual cortex. *CCN* 2026.
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